Jason Chen

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Education

University of Minnesota - Twin Cities — Minneapolis, MN
Bachelor of Science in Computer Science, 3.9 GPA, Dean's List (5 Semesters) — *September 2022 – December 2025*

- Relevant Coursework:
  Data Structures & Algorithms, Intro to Artificial Intelligence, Software Engineering
  Operating Systems, Machine Organization & Architecture, Program Design & Development

Experience

Teaching Assistant (Data Structures & Algorithms, Java) — Sept. 2023 – Present
University of Minnesota — *Minneapolis, MN*

- Collaborated with other TAs to lead and guide 50+ students through weekly labs
- Improved learning outcomes for 20+ students during weekly office hours by accommodating their learning styles
- Evaluated the projects of 50+ students using manual and automated JUnit testing

Projects

Drone Delivery Simulation | *C++, Git, Doxygen, Docker*

- Developed drone delivery simulation that handled 30+ deliveries via drone using search algorithms such as A*, BFS, DFS, and Dijkstra in C++
- Made use of Jira to assign and organize tasks working in a modified SCRUM environment
- Optimized drone pathing logic to reduce battery consumption by 18-22%

Spotify Playlists | *React, JavaScript, Git*

- Utilized Spotify's REST API to get songs, artists, recommendations, and playlist creation abilities
- Handles 10+ custom user inputs as seeds to generate a recommended playlist
- Directed users to log in to their account with Spotify's OAuth framework and add the generated playlist
- Implemented debounce to optimize input handling, leading to a 30% reduction of API calls

MyBudget App | *React, JavaScript, MongoDB, Python*

- Engineered a full-stack web application to automate the organization of bank transactions
- Integrated back-end logic with Python to read CSV data, reducing time spent on manual data entry by 80%
- Designed and constructed an API to store and edit over 100 persistent transactions with MongoDB

Multi Threaded TCP Server | *C*

- Designed and implemented a multi-threaded TCP server capable of handling concurrent HTTP read and write requests from multiple clients
- Utilized mutexes and condition variables to implement a thread-safe connection queue of clients

Face Mask Detector | *Python, TensorFlow*

- Trained a machine learning model with 99% accuracy utilizing TensorFlow that detects if a person is wearing a mask, wearing it incorrectly, or not wearing one
- Optimized the model's speed and accuracy by experimenting with multiple hyperparameters

League Data Tracker | *Python, TypeScript, Git, Angular*

- Retrieved and analyzed game performance metrics and statistics using the Riot Games API
- Developed a user-friendly front-end interface with TypeScript and Angular to display data clearly and comprehensibly

Path–Finding Visualization | *JavaScript, React, Git*

- Employed React to display a step-by-step view of the search path of multiple search algorithms
- Collaborated with a small team of developers, utilizing Git to implement features and resolve bugs
- Optimized search algorithms by testing multiple heuristics, resulting a 15% performance boost

Skills

Languages: Java, Python, C/C++, JavaScript, TypeScript, HTML/CSS, x86 Assembly, SQL, OCaml, UML
Frameworks: React, Node.js, Flask, Angular
Technologies: MongoDB, Docker, Git/Github, VS Code, IntelliJ, Doxygen